

Homework 3

CHAPTER 12

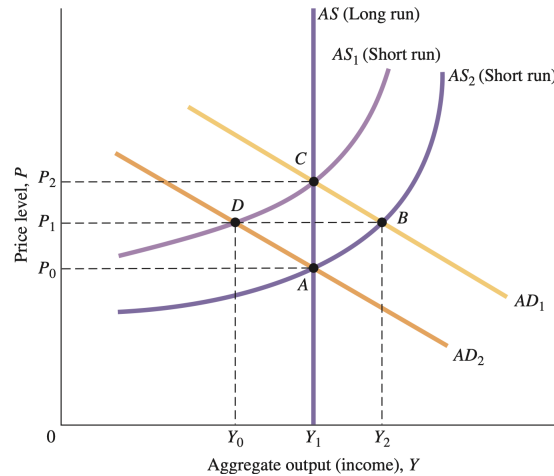
1.2. By using aggregate supply and aggregate demand curves to illustrate your points, discuss the impacts of the following events on the price level and on equilibrium GDP (Y) in the short run:

- (a) A tax cut holding government purchases constant with the economy operating well below full capacity
- (b) An increase in consumer confidence and business optimism with the economy operating at near full capacity

2.1. For each of the following scenarios, tell a story and predict the effects on the equilibrium level of aggregate output (Y) and the interest rate (r):

- (a) During 2009, the Federal Reserve was easing monetary policy in an attempt to boost the economy. That same year, Congress passed the American Recovery and Reinvestment Act which cut taxes and expanded existing tax credits for working families and businesses.
- (b) On January 1, 2018, the Tax Cuts and Job Act went into effect, lowering the top marginal tax rate for five of the seven income tax brackets. Assume that the Fed holds M^s fixed.
- (c) In 2014, the government raised taxes. At the same time, the Fed was pursuing an expansionary monetary policy.
- (d) In March 2018, consumer confidence in the United States rose to its highest level since 2007, reflected by a rise in consumption. Assume that the Fed holds the money supply constant.
- (e) The Fed attempts to increase the money supply to stimulate the economy, but plants are operating at 65 percent of their capacities and businesses are pessimistic about the future.

3.2. From the following graph, identify the initial equilibrium, the short-run equilibrium, and the long-run equilibrium based on the scenarios below. Explain your answers and identify what happened to the price level and aggregate output.



- Scenario 1. The economy is initially in long-run equilibrium at point A, and a cost shock causes cost-push inflation. The government reacts by implementing an expansionary fiscal policy.
- Scenario 2. The economy is initially in long-run equilibrium at point A, and an increase in government purchases causes demand-pull inflation. In the long run, wages respond to the inflation.
- Scenario 3. The economy is initially in long-run equilibrium at point C, and the federal government implements an increase in corporate taxes and personal income taxes. In the long run, firms and workers adjust to the new price level and costs adjust accordingly.
- Scenario 4. The economy is initially in long-run equilibrium at point C, and energy prices decrease significantly. The government reacts by implementing a contractionary fiscal policy.

CHAPTER 13

1.2. In 2019, the country of Sorbet was suffering from a period of high unemployment. The new president, Gelato, appointed Sherrie Sherbert as his chief economist. Ms. Sherbet and her staff estimated these supply and demand curves for labor from data obtained from the secretary of labor, Jerry Benjamin:

$$Q_D = 175 - 4W$$

$$Q_S = 16W - 30$$

where Q is the quantity of labor supplied/demanded in millions of workers and W is the wage rate in scoops, the currency of Sorbet.

- Currently, the law in Sorbet says that no workers shall be paid less than 12 scoops per hour. Estimate the quantity of labor supplied, the number of unemployed, and the unemployment rate.
- President Gelato, over the objection of Secretary Benjamin, has recommended to Congress that the law be changed to allow the wage rate to be determined in the market. If such a law was passed and the market adjusted quickly, what would happen to total employment, the size of the labor force, and the unemployment rate? Show the results graphically.
- Will the Sorbet labor market adjust quickly to such a change in the law? Why or why not?

CHAPTER 16

2.2. Tables 1, 2, and 3 that follow present some data on three hypothetical economies. Complete the tables by figuring the measured productivity of labor and the rate of output growth. What do the data tell you about the causes of economic growth? (*Hint:* How fast are L and K growing?)

TABLE 1

Period	L	K	Y	Y/L	Growth Rate of Output
1	1,120	3,205	4,650		
2	1,135	3,500	4,795		
3	1,152	3,798	4,945		
4	1,170	4,045	5,100		

TABLE 2

Period	L	K	Y	Y/L	Growth Rate of Output
1	1,120	3,205	4,650		
2	1,175	3,246	4,775		
3	1,255	3,288	4,904		
4	1,344	3,315	5,036		

TABLE 3

Period	L	K	Y	Y/L	Growth Rate of Output
1	1,120	3,205	4,650		
2	1,135	3,246	4,840		
3	1,152	3,288	5,038		
4	1,170	3,315	5,244		

CHAPTER 19

2.2. You are given the following model that describes the economy of Hypothetica.

- Consumption function: $C = 80 + 0.75Y_d$
 - Planned investment: $I = 49$
 - Government spending: $G = 60$
 - Exports: $EX = 20$
 - Imports: $IM = 0.05Y_d$
 - Disposable Income: $Y_d = Y - T$
 - Taxes: $T = 20$
- (a) Derive the planned aggregate expenditure function AE in this economy. Draw a graph showing AE and 45-degree line.
 - (b) Provide the equilibrium condition.
 - (c) What is the equilibrium income in Hypothetica? Denote the equilibrium point as E on the graph that you drew in part (a). What is the government deficit? What is the current account balance?
 - (d) If government spending is increased to $G = 75$, what happens to equilibrium income? Explain by using a graphical illustration which represents new AE as AE' and new equilibrium as point A on the graph that you drew in part (a). What happens to imports?
 - (e) Find the government spending multiplier.
 - (f) Now suppose the amount of imports is limited to $IM = 25$ by a quota on imports. If government spending is again increased from 60 to 75, what happens to equilibrium income? Explain why the same increase in G has a bigger effect on income in the second case. What is it about the presence of imports that changes the value of the multiplier?
 - (g) If exports are fixed at $EX = 20$, what must income be to ensure a current account balance of zero? (*Hint*: Imports depend on income, so what must income be for imports to be equal to exports?) By how much must we cut government spending to balance the current account? (*Hint*: Use your answer to be the first part of this question to determine how much of a decrease in income is needed. Then use the multiplier to calculate the decrease in G needed to reduce income by that amount.)

3.4. Suppose the exchange rate between the British pound and the U.S. dollar is $\text{£}1 = \$1.50$.

- (a) Draw a graph showing the demand and supply of pounds for dollars.
- (b) If the Bank of England implements a contractionary monetary policy, explain what will happen to the exchange rate between the pound and the dollar and show this on a graph. Has the dollar appreciated or depreciated relative to the pound? Explain.
- (c) If the U.S. government implements an expansionary fiscal policy, explain what will happen to the exchange rate between the pound and the dollar and show this on a graph. Has the dollar appreciated or depreciated relative to the pound? Explain.